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[22ME31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

MATERIAL SCIENCE & ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **5 X 2=10**

1.	a.	List any four properties of bio materials.	L1	CO1	02
	b.	Explain ultimate tensile strength.	L2	CO2	02
	c.	State Gibb's Phase rule.	L2	CO3	02
	d.	List out any four advantages of annealing.	L1	CO4	02
	e.	What are the piezoelectric materials? Give an example for the same.	L1	CO5	02
		PART-B	**6 X 5=30**		
2.	a.	Discuss the recent advances in smart materials for sensors and actuators application.	L2	CO1	05
		OR			
	b.	Find the effective number of atoms and atomic packing factor in the BCC unit cell.	L2	CO1	05
3.	a.	Discuss any five important applications of nano materials to medicine.	L2	CO1	05
		OR			
	b.	What is diffusion? Discuss the factors affecting diffusion in metals.	L1	CO1	05
4.	a.	What is a solid solution? Distinguish with a suitable examples substitution and interstitial solid solution.	L1	CO2	05
		OR			
	b.	Explain the different stages of creep with a neat sketch.	L2	CO2	05
5.	a.	Explain the Hume Rothary rules for the formation of substitution solid solution.	L2	CO3	05
		OR			
	b.	Explain any two types of corrosion.	L2	CO3	05
6.	a.	Discuss the various applications of aluminium alloys.	L2	CO4	05
		OR			
	b.	Describe flame hardening and carburizing heat treatment processes.	L2	CO4	05
7.	a.	Explain recent advances in magneto electric materials and its applications.	L2	CO5	05
		OR			

	b.	What are carbon nano tubes? Mention any three advantages and disadvantages.	L2	CO5	05
		PART-C			**6 X 10=60**
8.	a.	Explain different types of point defects.	L2	CO1	10
		OR			
	b.	Show that APF for FCC is 0.74	L3	CO1	10
9.	a.	Explain grain boundaries and tilt boundaries with a neat sketch.	L2	CO2	10
		OR			
	b.	Explain mechanism of diffusion with a neat sketch.	L2	CO1	10
10.	a.	Discuss the stages of cup and cone fracture with a neat sketches.	L2	CO2	10
		OR			
	b.	Discuss the types of fatigue loads with a neat sketches.	L2	CO2	10
11.	a.	Explain any two type's binary phase diagrams with a neat sketches.	L2	CO3	10
		OR			
	b.	Discuss any for types of corrosion prevention methods.	L2	CO3	10
12.	a.	Outline the important properties of cast iron. Explain the effect of various alloying element on cast irons.	L2	CO4	10
		OR			
	b.	Give the classification of Cu alloys. Discuss the different categories of brass along with their applications.	L2	CO4	10
13.	a.	Explain the concept of top-down and bottom-up approaches in nano technology. Discuss the recent applications of nano technology in environmental protection.	L2	CO5	10
		OR			
	b.	Explain in detail with examples nano materials of different sizes and shapes.	L2	CO5	10